Graphic Era

GEHU/02

End Term (Odd) Semester Examination December 2024

Roll no.

Name of the Course and semester: B.TECH. ME III SEM
Name of the Paper: THERMAL ENGINEERING
Paper Code: TME 308

Time: 3 hour

Maximum Marks: 100

Note:

(i) All the questions are compulsory.
(ii) Answer any two sub questions from a, b and c in each main question.
(iii) Total marks for each question is 20 (twenty).
(iv) Each sub-question carries 10 marks.

Q1. (2X10=20 Marks) CO 1

a. Describe the various types of boilers.

b. A gas of mass 1.5 kg undergoes a quasi-static expansion which follows a relationship $p = a + bV$, where a and b are constants. The initial and final pressures are 1000 kPa and 200 kPa respectively and the corresponding volumes are 0.20 m^3 and 1.20 m^3. The specific internal energy of the gas is given by the relation $u = 1.5\,pv - 85$ kJ/kg Where p is the kPa and v is in m3/kg Calculate the net heat transfer and the maximum internal energy of the gas attained during expansion.

c. A mass of gas is compressed in a quasi-static process from 80 kPa, 0.1 m3 to 0.4 MPa, 0.03 m^3. Assuming that the pressure and volume are related by pv^n = constant, find the work done by the gas system.

Q2. (2X10=20 Marks) CO 2

a. How nozzle and diffuser are the steady flow devices explain with the neat diagram.

b. Illustrate the equivalence of Kelvin Plank and Clausius Statement.

c. A turbine operates under steady flow conditions, receiving steam at the following state: Pressure 1.2 MPa, temperature 188°C, enthalpy 2785 kJ/kg, velocity 33.3 m/s and elevation 3 m. The steam leaves the turbine at the following state: Pressure 20 kPa, enthalpy 2512 kJ/kg, velocity 100 m/s, and elevation 0 m. Heat is lost to the surroundings at the rate of 0.29 kJ/s. If the rate of steam flow through the turbine is 0.42 kg/s, what is the power output of the turbine in kW.

Q3. (2X10=20 Marks) CO 3

a. Derive the Maxwell relations.

b. One kg of ice at -5°C is exposed to the atmosphere which is at 20°C. The ice melts and comes into thermal equilibrium with the atmosphere. (i) Determine the entropy increase of the universe (ii) What is the minimum amount of work necessary to convert the water back into ice at -5°C/ Cp of ice is 2.093 kJ/kgK and the latent heat of fusion of ice is 333.3 kJ/kg.

c. What is available energy and unavailable energy? What is the entropy expression for polytropic process?

Q4. (2X10=20 Marks) CO 4

a. Derive the efficiency of Diesel cycle.

b. Illustrate the factors affecting the efficiency of a Rankine cycle.

c. An engine operating on Otto cycle has a compression ratio of 6. The pressure and temperature at the beginning of compression are 1 bar and 27°C. Heat added during constant volume combustion process is 1170 kJ/kg. Determine the peak pressure and temperature, work output per kg of air and air standard efficiency.

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Q5. (2X10=20 Marks) CO 5

a. Illustrate the construction and working of a vapour absorption refrigeration

b. Define volumetric efficiency of a compressor. On what factors do it depends? Show that the optimum intermediate pressure of a two stage reciprocating compressor for minimum work is the geometric mean of the suction and discharge pressures.

c. A vapour compression refrigerator works between the pressure limits of 60 bar 25 bar. The working fluid is just dry at the end of compression and there is no under cooling of the liquid before the expansion valve.

Determine : 1. COP of the cycle 2. Capacity of the refrigerator if the fluid flow rate is 5 kg/min. Data:

Pressure (bar)	Saturation temperature, (K)	Enthalpy (kJ/kg)		Entropy (kJ/kg-K)	
		Liquid	Vapour	Liquid	Vapour
60	295	61.9	208.1	0.197	0.703
25	261	-18.4	234.5	-0.075	0.896